



8799 - SN Requiem and its Encore: Leveraging the first strongly lensed SN-Ia siblings for precision cosmology

Cycle: 4, Proposal Category: GO

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JWST Proposal 8799 (Created: Monday, October 20, 2025, 3:00:19PM Eastern Standard Time) - Overview

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	3		NIRCam Imaging	(1) Requiem2

ABSTRACT

We propose to obtain NIRCam imaging of the galaxy cluster that strongly lensed Supernova (SN) Requiem and SN Encore, enabling independent measurements of the Hubble constant (H_0) that are crucial given the current Hubble tension. The time delays between the appearances of the multiple images of a strongly lensed SN provide a direct measure of the time-delay distance and thus H_0 . SN Encore was discovered serendipitously in 2023 in a previous JWST program and is located close to the center of its bright host galaxy, making the photometry and time delay of SN Encore uncertain. SN Requiem was discovered in the same host galaxy in archival HST imaging taken in 2016, which showed three multiple images, and a fourth image of SN Requiem is expected to reappear in ~ 2026 based on our new cluster mass model. Our proposed NIRCam imaging has two goals: (1) Obtain straightforward template imaging for SN Encore, improving its photometry and thus the time-delay measurement of SN Encore by $\sim 50\%$, and (2) possibly detect the reappearance of SN Requiem with up to $\sim 18\%$ probability, enabling a $< 1\%$ precision measurement of its time delay. For only 2.5 hours, this program will make a competitive measurement of H_0 with SN Encore, and has the potential to measure H_0 from SN Encore and SN Requiem with $< 5\%$ uncertainty, an unprecedented precision from lensed SNe that paves the way for lensed SN cosmology.

OBSERVING DESCRIPTION

We propose a single visit of NIRCam imaging that matches the final epoch of SN Encore DDT observations (PID 6549). Our proposed observations will be used as a template for improved SN Encore photometry and will provide the first chance to detect SN Requiem reappearance. We repeat the final DDT observation without PA restrictions, though we do request scheduling in the latter two observing windows to increase the chance of detecting SN Requiem. We will optimize the pointing for the cluster once a scheduling window is assigned.

Proposal 8799 - Targets - SN Requiem and its Encore: Leveraging the first strongly lensed SN-Ia siblings for precision cosmology

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Requiem2	RA: 01 38 3.6020 (24.5150083d) Dec: -21 55 35.41 (-21.92650d) Equinox: J2000		
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO				

Proposal 8799 - Observation 3 - SN Requiem and its Encore: Leveraging the first strongly lensed SN-Ia siblings for precision cosmology

Observation	Proposal 8799, Observation 3									
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	Requiem2	RA: 01 38 3.6020 (24.5150083d)							
			Dec: -21 55 35.41 (-21.92650d)							
			Equinox: J2000							
	Comments: Category=Star Description=[Type Ia supernovae] Extended=NO									
Template	Module					Subarray				
	B					FULL				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX		4		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F115W	F444W	BRIGHT2	9	2	8	4	1589.042	
	2	F150W	F356W	BRIGHT2	6	2	8	4	1073.677	
	3	F200W	F277W	BRIGHT2	6	2	8	4	1073.677	
Special Requirements	After Date 20-DEC-2025:00:00:00									
	Offset 35.0 arcsec, 38.0 arcsec									